

SkyBook

An airline reservations platform, built end to end

Recruiter edition — an 11-page summary of the full engineering case study

Author: Praveenreddy Somireddy

At a glance

Item	Detail
What	A working airline-reservation platform: anonymous search, real seat maps, payment, confirmation email, check-in, scannable boarding pass, cancellations with fare-rule refunds, an admin console
Shape	Eight Spring Boot applications (one API gateway, seven domain services), React 19 web client, PostgreSQL, Kafka, full observability
Built	35 calendar days of designed increments, then a measured quality and release-engineering revision
Quality	SonarCloud gate PASSING: 91.2% coverage, 0 bugs, 0 vulnerabilities, security and reliability rating A
Tests	1,722 backend executions + 55 frontend + a 24-test end-to-end certification that races real buyers for one seat - zero failures at the evidence commit
Environments	A real promotion ladder: LOCAL > DEV > SIT > QA > PERF > UAT > STAGING > PROD, plus a scheduled weekly disaster-recovery drill - the first full destroy-and-restore cycle passed on 4 August 2026
Deployment	One cloud VM, 18 containers, automatic TLS - promoted by image digest, never rebuilt on the way to production
Author	Praveenreddy Somireddy - solo: requirements, architecture, backend, frontend, testing, CI/CD, cloud deployment and documentation, every decision and merge owned end to end

[Repository: github.com/praveencloudlab/skybook](https://github.com/praveencloudlab/skybook)

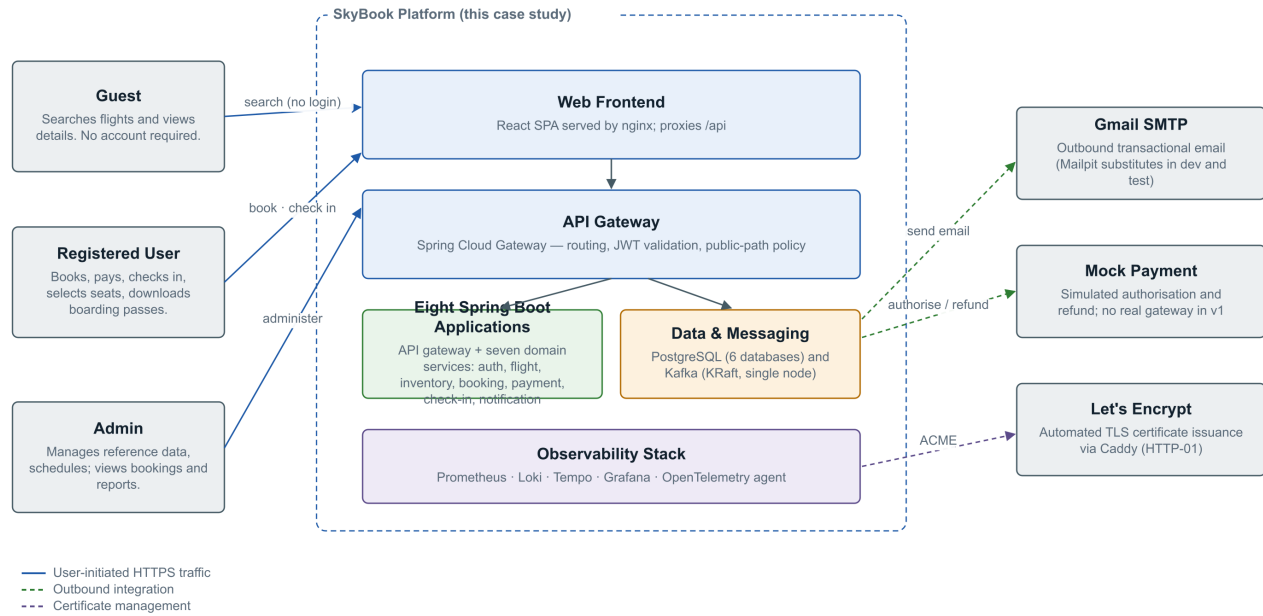
[Quality dashboard: sonarcloud.io/project/overview?id=praveencloudlab_skybook](https://sonarcloud.io/project/overview?id=praveencloudlab_skybook)

Disclaimer: SkyBook is a personal portfolio project - not a real airline, affiliated with no carrier. Airline-style codes in seed data are illustration only; payments are simulated server-side and no cardholder data exists; all passenger records are synthetic.

The system

Figure 1 — System Context

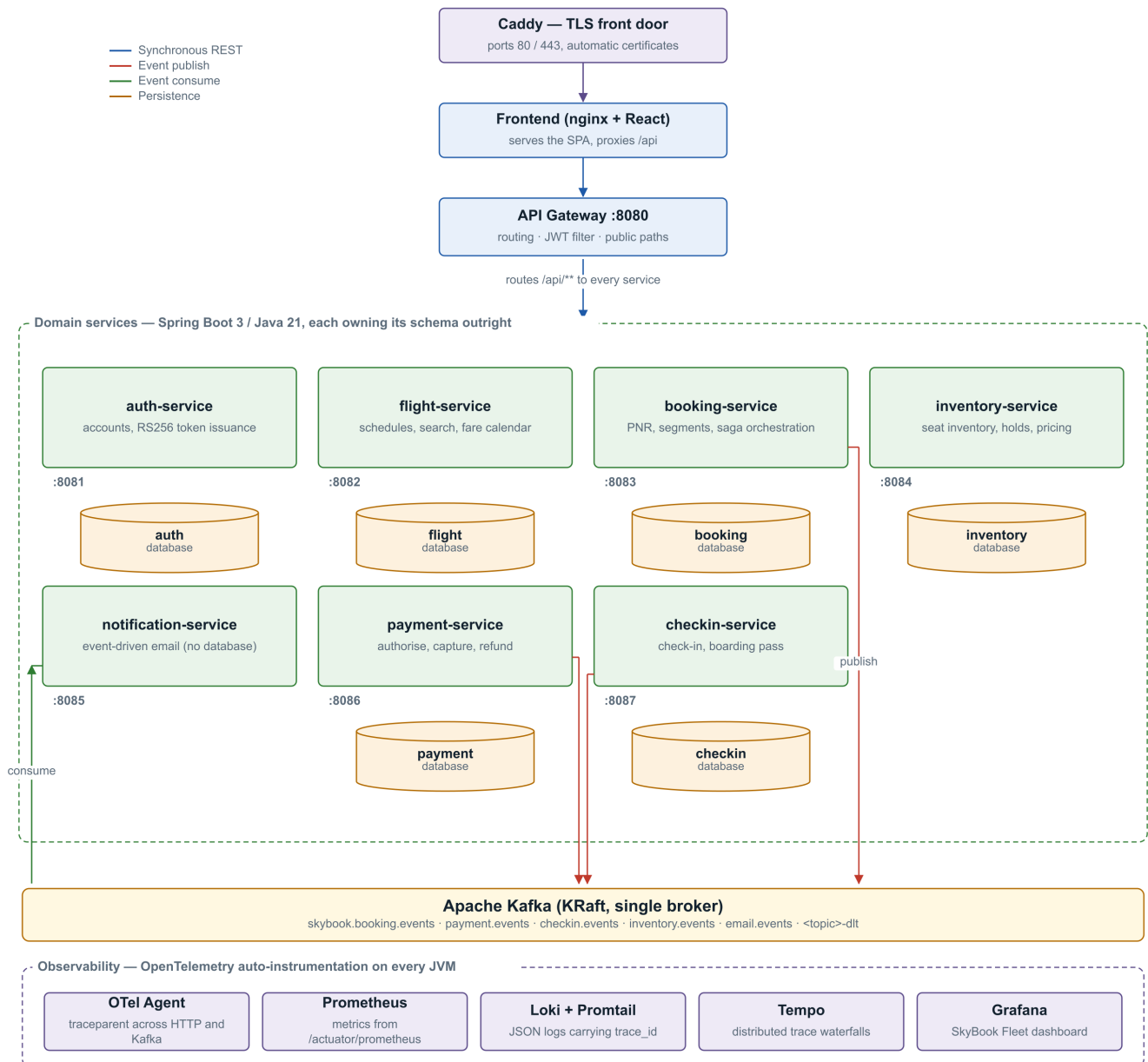
SkyBook passenger reservation platform and the parties around it



The context: a public shopping surface, an authenticated booking core, and operations behind an admin role.

Figure 2 — Microservice Architecture

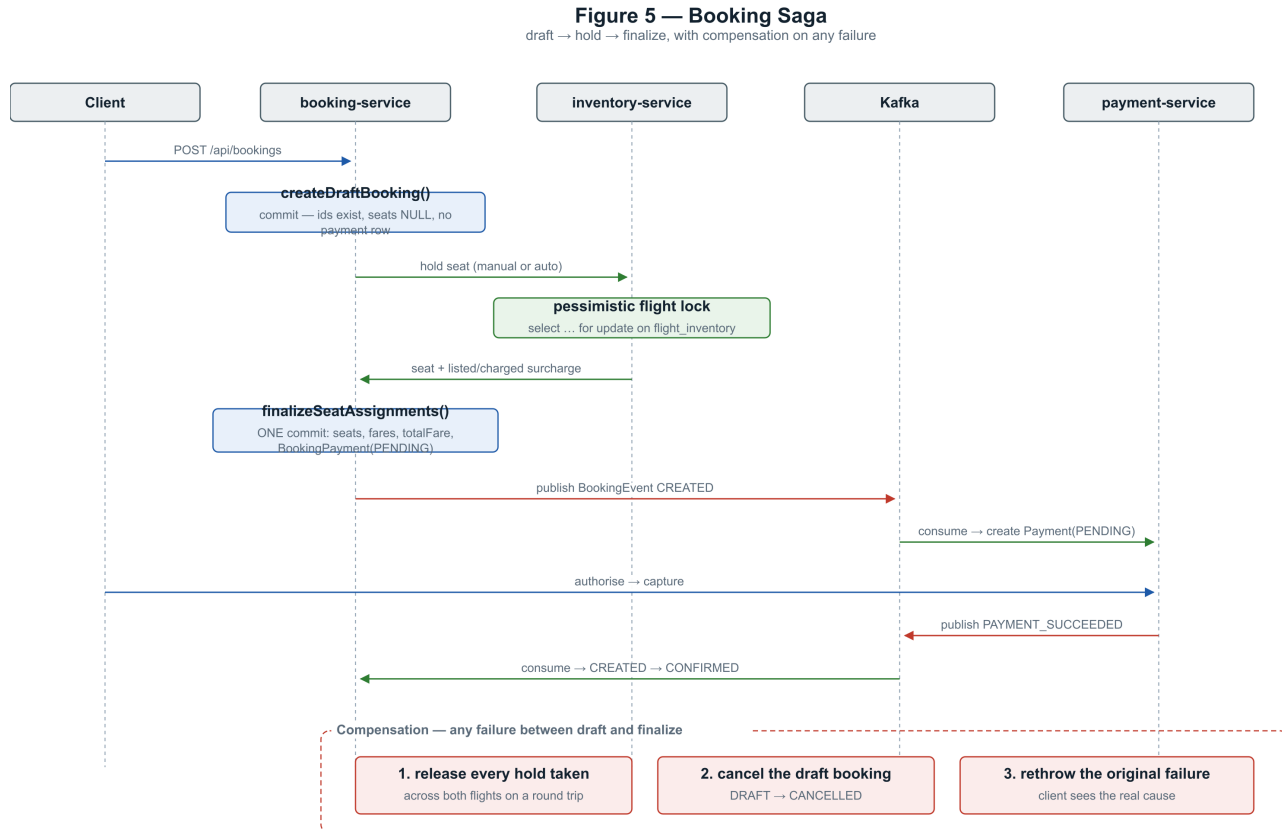
Runtime components, synchronous routes and asynchronous event flow



Eight applications over one PostgreSQL instance hosting six service-owned databases, and one event bus. Every service owns its data; nothing reaches another service's tables.

The hard part: money moves asynchronously

A booking is not confirmed when the payment call returns - it is confirmed when payment-service says so over Kafka and booking-service reacts. The saga below is the platform's spine, and the double-sell race (two real buyers, one seat, concurrent checkout) is certified end to end every night and on every promotion.



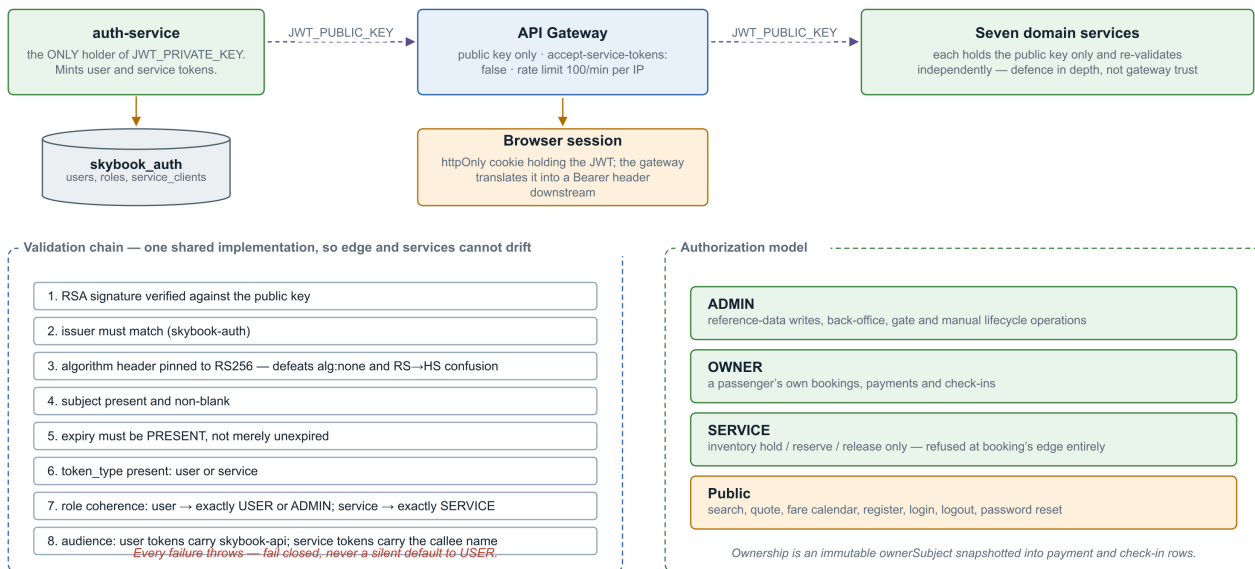
The facade is deliberately NOT @Transactional: every external call happens outside a database transaction, so no connection is held across network I/O.

The booking saga: draft, hold, pay, confirm - with compensations, not distributed locks.

Security, stated honestly

Figure 10 — Authentication and Authorization

RS256 asymmetric signing: one service mints, everyone verifies



RS256 sessions, per-service credentials, an httpOnly cookie the gateway exchanges for a Bearer header - and the gaps stated next to the controls.

The eight CSRF findings SonarCloud raised were closed with written justifications in code - the cookie is consumed at the gateway, SameSite=Lax covers the SPA, and all 48 GET endpoints were audited as read-only. A leaked mail credential early in the project was revoked, purged from history, and is recorded in the study as an incident, not hidden.

Evidence, not adjectives

Claim	Proof
1,722 backend test executions, zero failures	mvn clean verify at commit 96e24b2, 4 Aug 2026 - per-module table in the full study sums to exactly this
91.2% coverage, gate passing	Public SonarCloud dashboard, measured on every push
The journey certifies	24 e2e tests through the gateway only: booking, payment, check-in, boarding pass, double-sell race, email into a real sink, one trace across the Kafka hop
Fast enough under load	k6: 3,405/3,405 checks, p95 search 60 ms, quote 26 ms, 0.00% errors - thresholds gate every promotion
Backups actually restore	Scheduled weekly drill - first full cycle passed 4 Aug 2026: back up, destroy the database volume, restore, verify 1,336,748 rows against the manifest
Same artifact everywhere	Images built once per commit, scanned, published as multi-arch digests; every environment from DEV to PROD runs those exact bytes

The product

SkyBook English £ GBP Search flights Check-in Sign in

Search flights

A year of real schedules across 29 airports. Compare fares, pick your seat from the actual cabin — no account needed to look.

Round trip **One way** Multi-city

From **London Heathrow (LHR)** To **Dubai (DXB)** Guests and Cabin **1 Guest, Economy** Travelling when? **Sun 6 Sept** **Search**

3, Sept £93.50 4, Sept £102.85 5, Sept £93.50 **6, Sept £102.85** 7, Sept £93.50 8, Sept £93.50 9, Sept £93.50

Filters

STOPS

- ☒ Direct 4
- ☒ 1 stop 16
- ☐ 2 stops 0

SORT BY

Earliest Latest Shortest

DEPARTURE TIME

- ☒ Morning 05:00 – 11:59
- ☒ Afternoon 12:00 – 16:59
- ☒ Evening 17:00 – 20:59

20 trips LHR → DXB · Sun 6 Sept

Direct	6h 55m	from £102.85 per person	Select
08:25 LHR	6h 55m EK001	18:20 DXB	
06:53 LHR	7h SBZ713	16:53 DXB	
12:20 LHR	7h	22:20 DXB	

Search: a seven-day price strip and one card per itinerary - priced by the same endpoint checkout will use.

English

£ GBP

Admin console

Profile

Sign out

Booking in progress: LHR → DXB · 2026-08-17

Start a new search

Back

LHR → DXB

Mon 17 Aug · 08:25

Flights

Guests

Seats

Bags

Payment

Seat selection

Economy · Saver — secure your seat now, or let us assign one free at check-in.

Case Study

Free

Extra charge

Selected

Unavailable

Exit row

FRONT OF AIRCRAFT

15	A	B	C	D	E	F
16	A	B	C	D	E	F
17	A	B	C	D	E	F
18	A	B	C	D	E	F
19	A	B	C	D	E	F
20	A	B	C	D	E	F
21	A	B	C	D	E	F
22	A	B	C	D	E	F
23	A	B	C	D	E	F
24	A	B	C	D	E	F
25	A	B	C	D	E	F
26	A	B	C	D	E	F
27	A	B	C	D	E	F
28	A	B	C	D	E	F
29	A	B	C	D	E	F
30	A	B	C	D	E	F
31	A	B	C	D	E	F
32	A	B	C	D	E	F
33	A	B	C	D	E	F
34	A	B	C	D	E	F
35	A	B	C	D	E	F
36	A	B	C	D	E	F
37	A	B	C	D	E	F
38	A	B	C	D	E	F
39	A	B	C	D	E	F
40	A	B	C	D	E	F
41	A	B	C	D	E	F
42	A	B	C	D	E	F
43	A	B	C	D	E	F
44	A	B	C	D	E	F
45	A	B	C	D	E	F
46	A	B	C	D	E	F
47	A	B	C	D	E	F
48	A	B	C	D	E	F
49	A	B	C	D	E	F
50	A	B	C	D	E	F

REAR OF AIRCRAFT

No seat chosen

Skip — assign me a seat (free)

Back

Add seats later

Continue

Summary

Flight details

Mon 17 Aug

08:25

LHR

6h 55m · Direct

18:20

DXB

Economy Saver

London Heathrow to Dubai

Guests

Mr Case Study

Fare summary

Total:

Including taxes

£136.00

SkyBook

A working demonstration airline: real schedules, real seat maps, and a boarding pass you can actually scan.

OPERATIONS

Admin console

ACCOUNT

Signed in as **casestudy.de...**

Sign out

© 2025 SkyBook. A portfolio project — not a real airline.

29 airports · every pair served daily · a year of departures

Seat selection from the real cabin: paid rows priced per seat, held seats disabled, free auto-assign.

Page 8 of 11

 SkyBook

English

£ GBP

Admin console

Profile

Sign out



Payment received

Your booking reference

SBCXAT

Paid£152.30

Payment referencePAY-2026-E36Q49

Seat17B

StatusCONFIRMED

E-ticket

Case Study125-0000027601

Confirmed. A confirmation email is on its way.

Done



SkyBook

A working demonstration airline: real schedules, real seat maps, and a boarding pass you can actually scan.

OPERATIONS

Admin console

ACCOUNT

Signed in as casestudy.de...

Sign out

The moment three services agree: record locator, paid amount, trip saved.

[All trips](#)

BOOKING REFERENCE

SBCXAT

Booked Sun 16 Aug

Confirmed

£152.30

Download e-ticket

SB1906

Checked in

Mon 17 Aug

07:03

DXB

7h

Direct

11:03

MAN

Passenger (1)

Seats 17B

Case Study

Economy - Saver - seat 17B

£136.00

Total paid

£152.30

CONTACT

NAME

Case Study

EMAIL

casestudy.demo@skybook.test

PHONE

+447700900123

SkyBook

English

£ GBP

Admin console

Profile

Sign out

Case study

123-0000027001

Checked in

Manage booking

Cancellation & refund rules

- More than 72h before departure — full refund per your fare rules (Saver keeps its 30% fee; Flexi and Premium refund in full).
- 72h - 24h before departure — 50% of the fare-rule refund.
- Under 24h (same day) — cancellation still frees your seat, but the fare is not refunded.
- Last 2 hours before departure — online cancellation is closed; the airport desk can still help.
- Already checked in? — you can still cancel the whole booking; issued boarding passes are voided and the seats released.
- A refund due is returned automatically to the original payment method; check-in closes for every passenger.

Cancel booking...

Check-in & boarding passes

Case Study

SB1906 DXB → MAN Mon 17 Aug, 07:03 Seat 17B

Change seat

checked in

Download boarding pass

SkyBook

BOARDING PASS

ECONOMY

PASSENGER

CASE STUDY

BOOKING REF (PNR)

SBCXAT

DXB

Dubai - Terminal 1

MAN

Manchester - Terminal 2

FLIGHT

SB1906

DATE

17 Aug 2026

DEPARTS

07:03

BOARDING

06:18

TERMINAL

1

GATE

TBA

SEAT

17B

CABIN

Economy

BOARDING GROUP

4

NOTICE

Please arrive at the boarding gate by 05:48, 30 minutes before boarding begins at 06:18. The gate closes before departure and late passengers may be offloaded.

Issued 16 Aug 2026 13:32

PASSENGER

CASE STUDY

FLIGHT

SB1906

SEAT

17B

GROUP

4

BOARD

06:18

QR

BP-2026-AT9KLJ

PNR SBCXAT

SkyBook

A working demonstration airline: real schedules, real seat maps, and a boarding pass you can actually scan.

OPERATIONS

Admin console

ACCOUNT

Signed in as casestudy.de...

Sign out

© 2026 SkyBook. A portfolio project — not a real airline.

29 airports · every pair served daily · a year of departures

The boarding pass - the QR is real and the gate-operations console verifies it.

Page 10 of 11

Five things that went wrong, on purpose in print

Incident	Lesson
A transactional method silently ran with no transaction (Spring self-invocation)	Proxies do not apply to internal calls; the nightly schedule generator ran unprotected until static analysis flagged it
Every cross-timezone arrival displayed hours wrong - durations looked right by accident	Two wall clocks in different zones subtract to nonsense; fixing it touched generation, email, search, 414,679 rows and the screenshots
A round-trip cancellation could refund partially and leave the booking looking live	The expansion set and the remaining set were computed from different keys; pinned by tests that fail on the old code
The environment ladder's first ever run refused a bad promotion, then caught its own seeding bug	Gates that only pass are decoration; these blocked, and the fixes are commits
A test demanded zero collisions from a generator whose own name promised "rare"	The birthday paradox made CI red one run in twenty-two; assert what the code promises, not more

Current limits, stated: no transactional outbox (a post-commit publish failure loses the event - top of the roadmap), no sustained capacity test (the k6 gate catches regressions, not limits), single-VM deployment with RPO 24h / RTO ~30min, and a browser UI suite not yet wired into CI. The full study gives every gap its reasons.

The full engineering edition covers the architecture, the saga, pricing and refunds, check-in, the delivery history commit by commit, testing, defects and lessons, the data model, and six appendices - traceability, API contracts, security operations, screens, roadmap, glossary.